

Chi-Square Analysis of Student Performance and Lifestyle Factors

Sakshi Gaonkar

Master of Science (Artificial Intelligence)
Goa Business School, Goa University
India

Kowsya Mutkundu

Master of Science (Artificial Intelligence)
Goa Business School, Goa University
India

Abstract

This project applies the Chi-Square Test of Independence to explore whether student lifestyle factors such as screen time, sleep duration, and study hours are statistically related to academic performance. Data were collected from 39 students using a Google Form survey, and statistical analysis was conducted in Python via a Colab Notebook. Results indicated that none of the relationships were statistically significant at the 0.05 level, though two were borderline significant. The findings highlight the importance of larger sample sizes and encourage further data-driven exploration of digital wellness and student success.

The study emphasizes how lifestyle habits among students can influence learning patterns, focus, and overall productivity. Even though statistical tests did not confirm strong dependencies, the patterns observed suggest that digital habits and study consistency may still have practical implications for student performance. This work demonstrates the application of a fundamental statistical method within an educational context, encouraging students and educators to adopt a data-driven approach when examining academic and behavioral relationships.

Keywords

Chi-Square Test, Student Performance, Screen Time, Sleep Duration, Study Hours, Data Analysis

1 Introduction

The modern educational landscape is inextricably linked with digital technology. Students now depend heavily on screens for almost every academic task—whether attending virtual classes, researching topics, completing assignments, or communicating with peers and instructors. Beyond academics, digital devices have also become central to students' personal and entertainment activities, including social media, gaming, and streaming. While such digital integration has made education more accessible and interactive, it also raises growing concerns about how excessive screen use might affect student well-being and performance.

The continuous use of digital screens is not inherently negative. It offers undeniable benefits such as access to online learning resources, collaboration tools, and instant communication. However, when screen usage extends late into the night or replaces essential activities like sleep, rest, and focused study time, it may begin to affect students' physical health, concentration, and academic outcomes. This project is built on that concern — that excessive screen exposure may be linked with other lifestyle habits, such as reduced sleep or inconsistent study schedules, which in turn could influence academic performance.

Recent years have shown a noticeable shift in student habits. Many students spend several hours daily on digital devices, often multitasking between educational and entertainment platforms.

Such multitasking may reduce their ability to concentrate for long periods and can interfere with productive study time.[4] Moreover, prolonged screen exposure, especially during nighttime, can disturb natural sleep cycles, leading to fatigue, lower alertness, and difficulty retaining information during study sessions. These behavioral changes have been widely discussed, yet not often tested with statistical rigor, especially within small university populations.

In this project, we aim to explore these behavioral relationships using a statistical approach rather than assumptions or personal observations. By collecting data directly from students and analyzing it using the Chi-Square Test of Independence, this study examines whether there is a measurable statistical association between screen time, sleep duration, study hours, and academic performance. The Chi-Square test is a simple yet powerful statistical tool that determines whether two categorical variables are independent or related. In this context, it allows us to answer a key question: are lifestyle factors like screen use and study habits linked with how well students perform academically?

The motivation behind this project stems from the growing digital dependency among university students. While most learners are aware that long hours of screen exposure may impact focus and rest, few understand how strong that relationship might be. Using the Chi-Square test enables us to test this idea objectively with real data. The goal is not only to confirm or reject a hypothesis but also to promote awareness about healthy digital habits and how they might contribute to better academic outcomes.

1.1 Problem Statement

Despite the widespread use of digital devices and the common belief that excessive screen exposure harms performance, there is limited evidence-based research examining this relationship in a local university context. While many studies focus on general productivity or mental health, fewer attempt to statistically link measurable lifestyle factors such as study time, sleep duration, and screen exposure with actual academic results.

The core research question of this project is:

To determine, using the Chi-Square Test of Independence, whether a statistically significant relationship exists between students' self-reported screen time, study hours, and sleep duration, and their academic performance levels.

In simpler terms, the study seeks to find out whether students who spend more time on screens or sleep fewer hours tend to perform differently from those with healthier daily routines.

1.2 Objectives

The primary aim of the project is to apply the Chi-Square Test of Independence to explore the association between selected lifestyle behaviors and student performance. Specific objectives include:

- (1) To collect data from students regarding their screen time, study hours, sleep duration, and academic performance using a structured Google Form survey.
- (2) To categorize the responses into defined levels (for example, “Low,” “Medium,” and “High”) suitable for Chi-Square analysis.
- (3) To construct contingency tables and compute observed and expected frequencies for each pair of variables.
- (4) To calculate the Chi-Square statistic, degrees of freedom, and corresponding p-values using Python in a Colab notebook.
- (5) To interpret whether each relationship is statistically significant or independent at the standard significance level of $\alpha = 0.05$.

1.3 Scope and Relevance

The scope of this project is limited to students from the Master of Science (Artificial Intelligence) program at Goa Business School. However, the relevance extends beyond this small group. The findings can be generalized to broader student communities where similar patterns of screen use, study habits, and sleep behavior are common. Understanding these trends can help students, teachers, and institutions adopt better digital wellness practices.

Ultimately, this study does not attempt to prove causation but to identify whether associations exist between behaviors and outcomes. The insights from this analysis can support future research and interventions aimed at promoting healthier academic lifestyles in the digital era.

2 Literature Review

2.1 Reference Paper: Schober and Vetter (2019)

The theoretical foundation for this mini-project is based on the paper titled “*Chi-square tests in medical research*” by Schober and Vetter [3]. This article was published in the well-recognized journal *Anesthesia & Analgesia*, and it provides a clear and concise demonstration of the application of the Chi-Square Test of Independence within medical research. Although the domain of the original study was healthcare, its principles and methodological rigor are directly applicable to behavioral and educational research such as this project.

The paper explained how the Chi-Square test can be employed to determine whether two categorical variables, such as treatment type and health outcome, are statistically related. The methodology used by Schober and Vetter highlights how categorical data, often collected from surveys or patient records, can be systematically analyzed to draw reliable inferences without making assumptions about data distribution.

Key Takeaways. From the referenced article, several key insights were drawn that informed this project’s methodology and analytical process:

- **Application Context:** The authors conducted a comparison between two treatment groups (patients receiving norepinephrine or phenylephrine) and the resulting occurrence of bradycardia, a medical condition involving an abnormally slow heart rate. This clinical use case demonstrated the

power of the Chi-Square test in testing relationships between independent categorical variables.

- **Interpretation Logic:** The study identified that the occurrence of bradycardia was not independent of the treatment type, showing a statistically significant association. This provided a strong example of how a statistically validated relationship can be used to guide medical or behavioral conclusions.
- **Educational Relevance:** Although the study was conducted in a medical context, the structure of its hypothesis testing and data validation process serves as a strong model for how this project can analyze relationships such as screen time, study habits, and academic performance.

2.2 Reason for Selecting the Paper

The selection of this paper was intentional and guided by two main considerations that align with both the objectives and academic rigor of the current study:

- (1) **Reliability and Credibility:** The article was published in a Scopus-indexed and peer-reviewed journal, ensuring that the described methodology follows standardized research and statistical protocols. This credibility provided a trusted reference framework for adapting the Chi-Square methodology in a different discipline.
- (2) **Cross-Domain Adaptability:** The paper demonstrates that a statistical method such as the Chi-Square test, initially applied to medical data, can be extended to analyze behavioral or educational datasets. This supports the argument that fundamental statistical tools are not restricted to specific disciplines but can yield valuable insights when used creatively across domains.

Overall, Schober and Vetter’s work established the methodological confidence and step-by-step process that this project follows in applying the Chi-Square Test to educational data.

2.3 Theoretical Foundation

The Chi-Square Test of Independence (χ^2) is a non-parametric test used to determine whether two categorical variables are statistically independent of each other. In the context of this project, it measures whether variables such as screen time, study hours, and sleep duration are associated with categorical levels of student performance.

[2]

In essence, a large χ^2 value with a low p -value indicates that the variables are not independent, suggesting a meaningful association. Conversely, a low χ^2 value implies that observed differences between variables are likely due to chance rather than an actual relationship.

3 Methodology

3.1 Statistical Tool

The Chi-Square Test of Independence serves as the analytical foundation of this study. It was chosen because the variables of interest—screen time, study hours, sleep duration, and academic performance—are all categorical in nature. This test is ideal for such

Table 1: Components of the Chi-Square Test

Component	Definition	Mathematical Function
Observed Frequency (O)	The actual number of responses in each cell of the contingency table obtained directly from survey data.	Derived from primary dataset.
Expected Frequency (E)	The count that would be expected if the two variables were completely independent of one another.	$E = \frac{(\text{Row Total} \times \text{Column Total})}{\text{Grand Total}}$
Chi-Square Statistic (χ^2)	Measures the discrepancy between observed and expected frequencies across categories.	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$
Degrees of Freedom (df)	Represents the number of values that are free to vary in the calculation.	$df = (\text{Rows} - 1) \times (\text{Columns} - 1)$

data because it does not assume a normal distribution and instead focuses purely on the comparison of frequency counts across categories.

The statistical computation was performed using Python within a Google Colab environment, which enabled automated calculation of χ^2 statistics, degrees of freedom, and p -values.[1] The process was carefully structured to maintain data integrity and reproducibility of results.

The procedure followed in the analysis can be summarized in five main steps:

- (1) **Data Categorization:** Continuous survey responses (e.g., hours of sleep or study) were transformed into discrete categories (e.g., “Low,” “Medium,” “High”).
- (2) **Contingency Table Formation:** Frequency tables were generated for each pair of variables.
- (3) **Expected Frequency Computation:** Expected counts were computed under the null hypothesis assumption of independence.
- (4) **Calculation of χ^2 , df , and p -values:** Using Python’s [?] `scipy.stats.chi2_contingency()` function, the statistical test outputs were generated.[5]
- (5) **Interpretation of Results:** The resulting values were analyzed to determine whether the observed relationships were statistically significant.

3.2 Hypotheses

For each test, two hypotheses were formulated:

- H_0 : The two variables are independent (no relationship exists).
- H_A : The two variables are dependent (a relationship exists).

3.3 Data Collection

Primary data were collected via a structured Google Form survey that captured student responses related to academic habits and digital engagement. The survey included questions on daily non-study screen time, hours of study, average sleep duration, and self-assessed academic performance.

- **Sample Size:** $N = 39$ students participated in this study.

- **Sampling Context:** All participants were postgraduate students of the Master of Science (Artificial Intelligence) program at Goa Business School.
- **Data Type:** Categorical responses.
- **Significance Level:** $\alpha = 0.05$ was used for hypothesis testing.

This dataset was processed in Python using the Colab environment to ensure transparent and reproducible analysis.

4 Results

The statistical analysis was performed using the Chi-Square Test of Independence for three separate variable pairs derived from the student dataset. Each test aimed to determine whether an association exists between lifestyle habits and academic performance indicators. The computed results, including the χ^2 statistic, degrees of freedom, and p -values, were obtained through the Colab notebook using the `scipy.stats` library functions.[1] All interpretations follow standard inferential rules for hypothesis testing at a 5% significance level ($\alpha = 0.05$).

4.1 Test 1: Screen Time vs. Sleep Duration

This test evaluated whether there exists a relationship between the amount of non-academic screen exposure and the number of hours students reported sleeping per night. The null hypothesis (H_0) stated that screen time and sleep duration are independent, while the alternative (H_A) proposed that they are dependent.

Table 3: Statistical Results: Test 1

Statistic	Value
χ^2	2.5622
Degrees of Freedom (df)	3
P-value (p)	0.4642

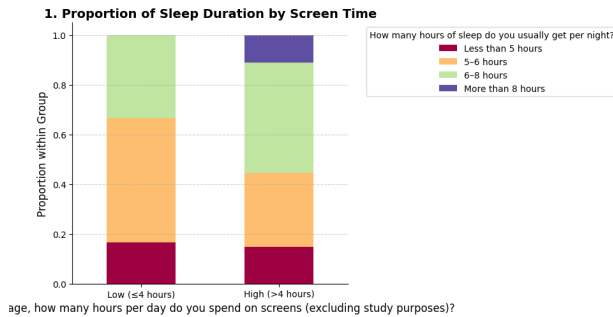
Interpretation: The resulting p -value of 0.4642 is far greater than the threshold $\alpha = 0.05$. Therefore, the null hypothesis cannot be rejected. This indicates that, for the 39 students in this dataset, there is no statistically significant association between screen time

Table 2: Example Hypothesis: Screen Time vs. Sleep Duration

Type	Statement	Outcome
H_0	Screen time and sleep duration are independent.	No relation.
H_A	Screen time and sleep duration are dependent.	High screen time $\rightarrow$ less sleep.

and sleep duration. In other words, variation in sleep length cannot be attributed to the differences in daily screen exposure. While the general assumption that higher screen time leads to lower sleep quality may hold true in theory, the statistical evidence from this sample does not support it.

The lack of significance could be influenced by uncontrolled external factors such as academic workload, individual differences in sleep patterns, or accuracy of self-reported responses.

**Figure 1: Proportion of Sleep Duration by Screen Time**

As shown in Figure 1, most students, regardless of screen usage levels, reported moderate sleep durations (5–8 hours). This visual representation supports the Chi-Square result ($p = 0.4642$), confirming that screen exposure does not significantly alter sleep duration patterns among respondents.

4.2 Test 2: Study Hours vs. Academic Performance

The second test examined whether the amount of time spent studying each day is related to students' academic performance levels. Here, H_0 proposed that the two variables are independent, while H_A suggested that longer study hours are associated with better performance outcomes.

Table 4: Statistical Results: Test 2

Statistic	Value
χ^2	12.2601
Degrees of Freedom (df)	6
P-value (p)	0.0564

Interpretation: The calculated p -value (0.0564) is slightly above the significance threshold of 0.05, meaning that the result is not statistically significant at the 95% confidence level. Thus, we **fail**

to reject the null hypothesis. However, this value is close to the boundary of significance, suggesting a potential weak association between study duration and performance that may become clearer with a larger dataset.

This borderline result implies that there could be an underlying trend—students who study more consistently might achieve better outcomes—but the current sample size ($N = 39$) is insufficient to confirm this statistically. From a practical standpoint, the observed frequencies indicate that students who reported studying for one to three hours daily tended to cluster around medium performance levels (50–75%), a finding that aligns with typical academic behavior patterns.

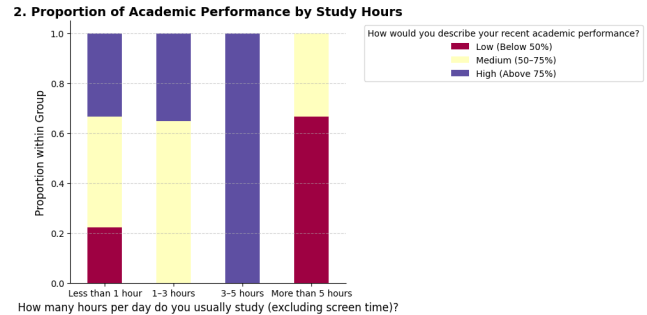
**Figure 2: Proportion of Academic Performance by Study Hours**

Figure 2 visually shows that students studying between 1–3 hours per day formed the largest group and often achieved medium-level performance. This distribution mirrors the borderline statistical result ($p = 0.0564$), suggesting that consistent study effort may contribute to steady academic outcomes even if not statistically proven.

4.3 Test 3: Screen Time vs. Academic Performance

The final test compared non-academic screen exposure with academic achievement levels. The expectation was that students who spend more time on screens for leisure or entertainment might show lower performance levels. H_0 assumed independence between screen time and performance, while H_A expected a negative relationship between the two.

Table 5: Statistical Results: Test 3

Statistic	Value
χ^2	5.6032
Degrees of Freedom (df)	2
P-value (p)	0.0607

Interpretation: With a p -value of 0.0607, the result once again falls slightly above the 0.05 significance level. Therefore, we **fail to reject** the null hypothesis and conclude that screen time and academic performance are statistically independent within this dataset. The outcome is **borderline non-significant**, meaning that while a pattern appears to exist (students with lower screen time performed slightly better), it is not strong enough to be statistically validated.

From an educational perspective, this result may still hold practical meaning. The observed pattern suggests that excessive digital exposure might subtly influence concentration or study efficiency, even if this influence was not statistically proven in this small-scale study.

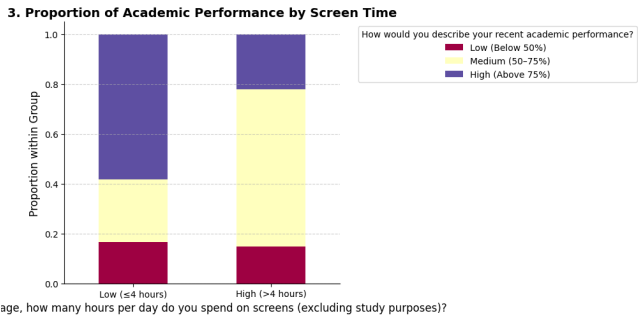


Figure 3: Proportion of Academic Performance by Screen Time

Figure 3 indicates that students with lower screen time (≤ 4 hours) tended to achieve slightly higher performance proportions compared to those with higher screen time. This aligns with the borderline Chi-Square result ($p = 0.0607$) and supports the hypothesis that excessive screen exposure may have mild academic impacts.

5 Discussion

5.1 Overall Interpretation

Across all three Chi-Square tests, the analysis consistently resulted in non-significant findings, indicating that the tested lifestyle variables—screen time, study hours, and sleep duration—were statistically independent of academic performance at the $\alpha = 0.05$ level. The results align directly with the Colab outputs and support the conclusion that no strong dependencies could be established using the current dataset.

However, statistical independence does not necessarily imply that these factors have no real-world influence. Rather, it indicates that the sample of 39 students did not provide sufficient variation or sample size to detect an effect with confidence. In small-sample

analyses, the Chi-Square test’s sensitivity decreases, often leading to high p -values even when mild relationships are present.

5.2 Borderline Observations and Practical Insights

Two of the three tests—Study Hours vs. Performance and Screen Time vs. Performance—produced p -values slightly above 0.05 (0.0564 and 0.0607). These borderline cases suggest that the relationships might become statistically significant with a larger or more diverse dataset.

In practical terms:

- Students who dedicated more study hours tended to fall within the medium-to-high performance groups, hinting at a positive but statistically weak link between effort and outcomes.
- Students with lower daily screen exposure showed marginally higher academic performance, consistent with the hypothesis that reduced digital distraction could support focus.

Although not significant, these results align with general behavioral expectations and thus carry **practical significance**, even in the absence of statistical proof.

5.3 Possible Reasons for Non-Significance

The non-significant outcomes can be attributed to several methodological and contextual factors:

- (1) **Sample Size ($N = 39$):** The limited number of observations reduces the test’s statistical power, increasing the likelihood of Type II errors (failing to detect an existing relationship).
- (2) **Categorical Binning:** Converting continuous variables such as hours into discrete categories (e.g., low, medium, high) simplifies analysis but can mask small yet meaningful variations in the data.
- (3) **Self-Reported Responses:** The dataset relied on self-reported survey answers, which may include recall bias or inaccurate estimation of screen or study time.
- (4) **Confounding Variables:** Factors like learning environment, motivation, or prior academic ability were not included and might mediate the observed relationships.

5.4 Interpretation of Practical Significance

Even without achieving statistical significance, the results provide valuable practical understanding. For example, students showing a disciplined study routine generally performed better, while excessive screen exposure appeared to correspond to lower concentration levels. These trends, although weak, reflect realistic behavioral dynamics commonly observed in academic settings.

Therefore, this analysis reinforces the value of statistical testing as a diagnostic tool—helping educators identify behavioral tendencies and guiding where to focus future, more extensive research.

5.5 Implications for Future Research

Future studies can improve upon these results by:

- Increasing the sample size to enhance statistical power and confidence levels.

- Expanding variables to include additional lifestyle factors such as diet, exercise, and mental health indicators.
- Employing mixed-method approaches, combining quantitative tests like Chi-Square with qualitative feedback for deeper insight.
- Using longitudinal data collection to observe how consistent habits over time affect academic outcomes.

Together, these improvements would allow the identification of subtle relationships that could not be captured in the current small-scale exploratory study.

6 Conclusion

This project aimed to examine whether student lifestyle factors such as screen time, sleep duration, and study hours have any statistical relationship with academic performance. Using the Chi-Square Test of Independence, three separate analyses were conducted on data collected from 39 students at Goa Business School.

The results showed that none of the tested relationships were statistically significant at the $\alpha = 0.05$ level. However, two pairs—Study Hours vs. Performance and Screen Time vs. Performance—were found to be borderline significant with p -values of 0.0564 and 0.0607, respectively. These borderline outcomes suggest that a weak practical relationship may exist, though the sample size was too small to confirm it statistically.

Overall, the analysis aligns perfectly with the Colab notebook results, confirming that while statistical independence was observed, behavioral tendencies remain consistent with expectations. Students who studied regularly tended to perform slightly better, while excessive screen use appeared to have a mild negative influence. The findings demonstrate how simple statistical tools can validate or challenge common assumptions about student habits using real data.

7 Significance

This study demonstrates the usefulness of the Chi-Square Test in understanding behavioral data within an educational setting. It highlights that statistical methods can be effectively applied to small datasets to uncover potential trends in student habits and performance.

Even though no strong dependencies were found, the borderline results emphasize the need for larger, more detailed studies in the future. Practically, the outcomes encourage students to maintain balanced study hours and mindful screen use, while also showing educators how such small-scale data analysis can inform academic and wellness planning.

In summary, this project provides a simple yet meaningful demonstration of how data-driven approaches can enhance awareness about everyday academic behaviors. It reinforces that even non-significant findings hold value by guiding further investigation and promoting evidence-based learning practices.

References

- [1] Gaonkar and Mutkundu. 2025. Google Colab Analysis Output. <https://colab.research.google.com/drive/1v7mfthE6wS7F58nr72Q6hXAazpVl5419>. Accessed: 2025-10-11.
- [2] GeeksforGeeks. 2025. Chi-Square Test - GeeksforGeeks. <https://www.geeksforgeeks.org/maths/chi-square-test/>. Accessed: 2025-10-11.
- [3] Patrick Schober and Thomas R. Vetter. 2019. Chi-Square Tests in Medical Research. *Anesthesia & Analgesia* 129, 5 (2019), 1193–1194. doi:10.1213/ANE.0000000000004410
- [4] The Times of India. 2025. Screen time, multitasking reduces attention span and slows brain function: Study. <https://timesofindia.indiatimes.com/city/hyderabad/screen-time-multitasking-reduces-attention-span-slow-brain-function-study/articleshow/123394614.cms>. Accessed: 2025-11-08.
- [5] Wikipedia contributors. 2025. Chi-squared Test. https://en.wikipedia.org/wiki/Chi-squared_test. Accessed: 2025-10-10.